DualBalance Districting: From Detection to Generation

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Abstract

Partisan gerrymandering is structurally entrenched, and detecting it after the fact (the goal of most forensic metrics) cannot substitute for preventing it, particularly after *Rucho v. Common Cause* withdrew federal courts from partisan-gerrymandering review. This paper asks what would happen if congressional districts were drawn not by legislators but by a fully specified mathematical rule, with no human discretion at any stage. The candidate rule, *DualBalance Districting*, computes districts as a pure function of state geometry, census population, and seat count N, with every design choice pre-committed before any state's data are examined; each district carries roughly $1/N$ of the state's people and $1/N$ of its land, weighted equally in the DualBalance Score.

Applied to all 40 states with available VTD data, DualBalance achieves *Karcher*-compliant population balance on 28 states at VTD scale—the same granularity at which Census data are delivered, and coarser than the block-level precision conventionally required to satisfy the congressional equality standard—while both deterministic baselines achieve it on zero states. This comes at the cost of substantially lower compactness than enacted plans (median Polsby-Popper 0.115 vs. 0.281), a structural consequence of spanning the full urban-rural gradient in each district. A rotation sensitivity analysis across 12 anchor angles per state finds the DualBalance Score stable (cross-state median $\sigma = 0.010$) while projected seat counts shift by 1–3 seats in 25 of 40 states, identifying seed rotation as a consequential pre-committed choice that any legislative adoption would need to specify.

The paper's contribution is to demonstrate feasibility and document the properties of an apolitical deterministic design; whether such a rule constitutes fair representation is a normative question the algorithm cannot answer, and that judgment belongs to legislatures, courts, and voters.

1 Introduction

In most U.S. states, congressional district boundaries are drawn by the same legislators whose electoral prospects depend on where those boundaries fall. This structural conflict of interest is not resolved by the Constitution, which mandates only that House districts achieve population balance (Art. I, §2) while leaving every other line-drawing decision to political actors. The consequences are well-documented: gerrymandered boundaries reduce electoral competition, dilute minority voting power, and produce seat allocations that diverge substantially from statewide vote totals [2,16]. If a procedural alternative is to succeed, it must navigate three distinct challenges: the shrinking legal landscape for judicial oversight (§1.1), the geographic reality of what equal representation requires within a state (§1.2), and the backward-looking nature of the existing measurement toolbox (§1.3). This paper investigates whether a single deterministic rule can satisfy all three at once.

1.1 The Legal Landscape

The federal courts have largely withdrawn from gerrymandering review, in two stages. Section 2 of the Voting Rights Act, as construed in *Thornburg v. Gingles* [19], established the primary federal channel for challenging discriminatory maps: where a minority community is sufficiently large and compact, politically cohesive, and subject to majority bloc voting, majority-minority districts are required. For decades, that racial channel coexisted with the possibility of partisan-gerrymandering claims. In *Rucho v. Common Cause* [22], the Supreme Court closed the partisan channel entirely, holding 5–4 that such claims present nonjusticiable political questions; the Court conceded the practice is "incompatible with democratic principles" but found no manageable judicial standard. After *Rucho*, the *Gingles* framework remained the only federal route for gerrymandering review. That channel has since narrowed as well. *Allen v. Milligan* [23] reaffirmed in 2023 that Section 2 requires majority-minority districts (MMDs) where the *Gingles* preconditions are met. One year later, *Alexander v. South Carolina Conf. of the NAACP* [25] raised plaintiffs' evidentiary burden; *Louisiana v. Callais* [26] held in April 2026 that even a plan drawn to comply with a prior Section 2 order may violate the Equal Protection Clause if Section 2 did not in fact compel a race-based remedy. Together, *Alexander* and *Callais* leave race-conscious line-drawing in a narrower window of constitutional safety than at any point since the VRA's adoption.

State constitutional review under *Moore v. Harper* [24] remains available but is uneven and politically contingent: roughly ten state supreme courts have recognized state-constitutional partisan-gerrymandering claims [11]; the rest have not. Several states have already redrawn maps mid-decade in response to electoral results rather than census revision, and calls for an explicit redistricting war to maximize partisan advantage have brought the practice into open national political strategy [8]. Redistricting has shifted from a once-per-decade event into a recurring instrument of partisan competition. Any procedural alternative must therefore operate without relying on legal channels that are no longer available, and without producing inputs that trigger the narrowing doctrines described above.

1.2 Geography as the Instrument of Manipulation

Geography is the mechanism of partisan gerrymandering. When legislators draw district lines, they exploit the geographic sorting of partisan voters: packing opposition supporters into a small number of districts wastes their votes by producing unnecessary supermajorities, while cracking their communities across multiple districts wastes their votes by keeping them in permanent minority. No party affiliation need appear in the line-drawing software. The line-drawer needs only to know where each party's voters live, and geometry does the rest [2]. The result is a structural advantage that can persist for a decade regardless of shifts in public opinion.

If geography is the instrument of manipulation, it can also be the basis of a principled rule. Any procedural alternative must choose, explicitly and in advance, what a district is *for*. Current law answers with a single quantity: population. *Wesberry v. Sanders* [17] established that congressional districts must hold roughly equal shares of people, and for sixty years that has been the only legally required design criterion.

This paper proposes adding a second criterion: each district should also carry roughly $1/N$ of the state's geographic area. That is a radical departure from current law, and the motivation requires care. The argument is not that land votes. It is that when districts optimize only for population, they naturally cluster within the state's densest areas. In a heavily urbanized state, all districts can plausibly sit inside a handful of metropolitan counties, producing representatives who have no rural or exurban constituents and no practical reason to engage with them. The rural population becomes an afterthought of every district rather than the primary concern of any. Requiring each district to also carry a share of the state's geographic extent forces every district to span the full density gradient, from urban core to rural boundary, so that every representative is accountable to both. The question is not whether land votes; it is whether rural and urban populations alike deserve representatives who must engage with both.

The need is empirically grounded. American voters have increasingly sorted geographically by political values, clustering with others who share their outlook in ways that have intensified over the past generation [1]. Urban and rural communities differ systematically not only in partisan affiliation but in underlying policy priorities: land use, infrastructure, the role of government services, and the relationship between local economies and federal policy are experienced very differently at different population densities [14]. When districts are drawn to contain only one type of community, representatives have no electoral incentive to engage with the other. The geographic concentration of urban voters in particular has produced structural partisan asymmetries that persist regardless of statewide vote totals [2]. A district drawn to span both environments does not eliminate these differences, but it does require the representative to be accountable to constituents who hold them.

That framing does not resolve every objection. The one-person-one-vote principle is satisfied by *Karcher* compliance: every district holds an equal share of the population, and every person's vote carries identical population weight. Area balance is layered on top of that constraint, not in place of it. The more precise objection is normative rather than constitutional: adding geographic area as a second criterion encodes a preference for distributing districts across the density landscape, which gives communities spread across large areas a structural feature that dense communities do not share. Whether that is a desirable feature of a representative system, or an unjustified advantage for lower-density communities, is a question this paper does not resolve.

The case for investigating it rests on three observations. First, the constitutional structure has never treated population as the only legitimate dimension of representation: the Senate apportions by state regardless of population (Art. I, §3); Madison's defense in *Federalist* Nos. 54–58 [7] frames bicameralism as a principled refusal to collapse representation onto a single dimension. Second, a district that must span the full density gradient cannot be deliberately packed into or cracked across a single demographic community, which limits the geographic levers available to a line-drawer. Third, the equal-area criterion is *pre-committed*: fixed before any state's data are examined, it cannot be tuned to favor one community over another in a given cycle.

We adopt this as a working assumption, test whether a rule built on it is technically achievable and legally characterizable, and report honestly what it produces. Whether it should be adopted is a question for legislatures and voters to answer.

A typical U.S. state varies by two to three orders of magnitude between its densest census tract and its sparsest rural block. Equal population, equal area, contiguity, and shape compactness cannot all be satisfied simultaneously; some trade-off is unavoidable. Whatever criterion a rule adopts, it must commit to that trade-off explicitly and in advance, before any specific state's geometry is examined.

1.3 The Forensic Toolbox and Its Limits

Quantitative tests for gerrymandering include shape-compactness measures (Polsby-Popper [12]; Reock [13]), partisan-asymmetry statistics [31,32], the Efficiency Gap [16], and ensemble outlier methods [4,5]. All of these are *forensic* instruments: they take an enacted map as input and ask whether its observed properties are consistent with maps that some neutral reference process would have produced absent partisan intent. They compare an enacted plan against a counterfactual to infer the line-drawer's motive, and were built for an adversarial context in which the question is whether somebody did something they should not have done.

A generative criterion is a different object. It states, directly, what a district *should* look like, and is used as the objective of the line-drawing procedure rather than as evidence about the procedure. Population balance is the only generative criterion U.S. redistricting law currently uses: *Wesberry v. Sanders* [17] and its congressional-district elaboration in *Karcher v. Daggett* [18] require that each congressional district hold as nearly equal a share of the population as practicable. Every other criterion (compactness, communities of interest, partisan fairness) enters the law as a forensic instrument or a discretionary guideline.

Forensic metrics also presuppose a line-drawer: the inference from a plan's shape or partisan asymmetry to manipulative intent requires that somebody exercised discretion that could, in principle, have gone differently. A procedure that eliminates that discretion severs the inference chain at its foundation. What is needed is not a better forensic test but a different kind of rule entirely, one that specifies what districts should look like rather than evaluating whether they were distorted.

1.4 A Candidate Procedural Framework

Iowa's Legislative Services Agency [6] has operated under a written, publicly documented redistricting procedure since 1980, one of the few U.S. examples of formalized algorithmic redistricting that has functioned without major legal challenge across multiple cycles. Its longevity suggests that rule-based redistricting can be politically sustainable. The limiting question is whether a procedure designed for Iowa's relatively compact, monocentric geography generalizes to states with substantially different population distributions and geographic structures. We re-implement the Iowa formula as a structured baseline (referred to here as *Cascade* to avoid confusion with the agency) and use it as a point of comparison. Cascade captures the geographic-prioritization logic of the Iowa procedure (county integrity first, then population balance, then compactness) but omits the institutional framework (the bipartisan advisory commission, multiple plan submissions, and legislative up-or-down vote) integral to Iowa's actual process; the comparison therefore isolates the algorithmic contribution of county-preservation from the process surrounding it.

The candidate we investigate beyond Iowa is *DualBalance Districting*: a deterministic algorithm whose output is a pure function of state geometry, census populations, and seat count. It is designed to engage each of the three challenges identified above.

Against the *legal challenge*: every design choice is pre-committed before any state's data are examined, producing a rule with no within-state discretionary parameters and no partisan, racial, or incumbent inputs. The intent-attribution chain that forensic metrics rely on is severed because there is no line-drawer exercising discretion within the state.

Against the *geographic challenge*: each district explicitly carries both $1/N$ of the state's people and $1/N$ of its land, weighted equally. A radial seeding geometry naturally spans the full density gradient from urban core to rural boundary in each district, making the geographic trade-off a first-class design commitment rather than an afterthought.

Against the *forensic limitation*: the DualBalance Score is a *generative* criterion. It states what a district should be, one that represents both a people-share and a land-share, and serves as the objective the algorithm pursues rather than a test applied after the fact. Conventional metrics (Polsby-Popper, Efficiency Gap, majority-minority counts) are computed and reported, but as descriptions of the partition, not as the basis for design.

Whether this candidate succeeds, and where it falls short, is what this paper documents.

2 Methods

2.1 Data and inputs

Atomic units are U.S. Census Bureau *voting tabulation districts* (VTDs) and *census tabulation blocks* from the 2020 decennial release [29], obtained via TIGER/Line shapefiles [28]. Boundaries are used as-is; no simplification or smoothing is applied. Population counts are the total-population figures from PL 94-171. The enacted congressional plan used for comparison is the 119th-Congress plan (effective January 2025), joined to VTDs and blocks via a spatial representative-point join against the TIGER 2024 `cd119` layer [30].

Note on enacted-plan population deviations. Enacted U.S. congressional plans are litigated to *Karcher* compliance at the census-block level. The enacted-plan deviations reported in this paper reflect the VTD-layer spatial join: a VTD that straddles a district boundary is assigned in full to one district, introducing apparent deviation where the legislative record shows none. DualBalance's reported deviations come from its own block-refined output and are directly comparable to the *Karcher* threshold. Counts of enacted-plan Karcher compliance (Tables 1–2) should be read as a conservative lower bound on enacted compliance, not as legally cognizable deviations: the true enacted deviations are near zero on block-accurate data.

Reproducibility. The `dualbalance` Python package and per-state configuration files reproduce all tables and figures from publicly available TIGER and Census source files. Algorithm outputs are byte-identical across repeated runs on the same input data; the `votes_source` field in each state's units file records which election data source was used.

2.2 Problem statement

Given a state S partitioned into atomic units $U = \{u_1, \ldots, u_M\}$ (census tabulation blocks, block groups, or voting tabulation districts (VTDs), in increasing order of size) and a target district count N, DualBalance Districting produces a deterministic assignment $\pi : U \to \{1, \ldots, N\}$ such that each district $D_i = \pi^{-1}(i)$ is contiguous, non-empty, and as close as the geometry allows to representing both $1/N$ of the state's people and $1/N$ of its geography. Let $P = \sum_u \mathrm{pop}(u)$ and $A = \sum_u \mathrm{area}(u)$ with per-district targets $P^* = P/N$ and $A^* = A/N$.

The full pipeline runs in three stages: a radial-seed stage, *DualBalance* (§§2.3–2.5), which provides the deterministic starting configuration; a VTD-scale tightening stage (§2.7) which drives $\max_i |\delta_i|/P^*$ toward the user-supplied tolerance τ via a hybrid greedy local search with bounded-chain escape; and a block-scale refinement stage (§2.8) which re-initializes at finer granularity to recover area balance within the achieved pop-deviation envelope. Each stage is a pure function of its inputs.

2.3 Radial seed placement

Compute the population-weighted centroid

$$c = (c_x, c_y) = \left(\frac{\sum_u p_u x_u}{\sum_u p_u}, \frac{\sum_u p_u y_u}{\sum_u p_u} \right),$$

where x_u, y_u are the centroid coordinates of unit u in an equal-area projection and p_u its population. Let diag denote the bounding-box diagonal of the units and set the seed radius $r = 0.001 \cdot \text{diag}$. For $d = 0, 1, \ldots, N-1$ place seed s_d at

$$s_d = \big(c_x + r\cos(2\pi d/N),\; c_y + r\sin(2\pi d/N)\big).$$

Seed 0 sits due east of the centroid; seeds advance counter-clockwise by equal angular steps. The choice $r = 0.001 \cdot \text{diag}$ is small enough that the Voronoi cells generated by the seeds degenerate to near-perfect radial slices through c, but large enough to keep the seed positions numerically distinct.

These angular parameters, seed 0 due east and counter-clockwise advance, are pre-committed design choices fixed before any state-specific data are examined. Different anchor angles $\theta \in [0,\, 2\pi/N)$ produce different partitions with different partisan and demographic properties, because the resulting radial slices land differently on the state's population geography. Rotation sensitivity, i.e., the distribution of DBS, |EG|, and seat outcomes across the full rotation range, is therefore an important robustness check; results are reported in §3.1. The due-east anchor was selected as the natural zero of the standard angular convention; an adopting body would need to pre-specify a rotation rule (or adopt this one) as part of any legislative implementation.

The radial configuration is what carries the dual-balance property: each slice spans both the dense (near-c) and sparse (boundary-side) territory of the state, so the population it inherits is bounded by the cap (§2.4) while the area it inherits is driven toward A^* by the slicing geometry.

2.4 Capacitated first-fit assignment

Let $d(u, i) = \|x_u - s_i\|$ be the Euclidean distance from unit u to seed i. Initialize remaining capacities $\rho_i = P^*$ for $i = 1, \ldots, N$. Sort all (u, i) pairs by ascending normalized distance $d(u, i)/\text{diag}$ and walk the sorted list in order:

for each (u, i) in ascending $d(u, i)/\text{diag}$:
  if u already assigned: continue
  if $\rho_i \geq p_u$: assign u to i; $\rho_i \leftarrow \rho_i - p_u$
  else: skip

Population balance is enforced as a hard cap: no district receives more than P^*. Any unit not placed by the end of the walk (a rare integer-rounding edge case) is assigned to the district with the largest remaining capacity; `argmax` resolves ties to the smallest district id.

Ties in normalized distance break by ascending (unit_id, district_id). There is no Lloyd recentering, no iteration count, no convergence test: the radial seeds do not drift, so a single assignment pass suffices.

2.5 Contiguity repair

After §2.4 every unit belongs to exactly one district, but a district may consist of more than one connected component (rare on convex states; more common on those with peninsulas, islands, or rural enclaves). For each such district, the largest connected component by total population is retained; units in the smaller components are reassigned to neighboring districts one at a time, in the lowest-cost direction. Cost is

$$c(u, j) = \frac{d(x_u, s_j)}{\text{diag}} + \frac{|P(D_j) + p_u - P^*|}{P^*} + \frac{|A(D_j) + a_u - A^*|}{A^*},$$

combining a normalized distance term with normalized population and area deviation penalties for the receiving district. Ties break in cascade $(c, \text{pop\_pen}, \text{area\_pen}, \text{dist}, \text{district\_id})$ ascending. The repair sweep iterates until no district has more than one connected component, capped at ten sweeps; in practice it converges in zero or one sweep on real census geometries.

Algorithm 1 summarizes the full core pipeline.

2.6 Scoring

Define per-district relative deviations $\text{pop\_dev}_i = |P(D_i) - P^*|/P^*$ and $\text{area\_dev}_i = |A(D_i) - A^*|/A^*$ and let $\overline{\text{pop\_dev}}$, $\overline{\text{area\_dev}}$ denote their means over $i = 1, \ldots, N$. The DualBalance Score is

$$\text{DBS} = \frac{1}{1 + \frac{1}{2}\,\overline{\text{pop\_dev}} + \frac{1}{2}\,\overline{\text{area\_dev}}}. \qquad (1)$$

Algorithm 1 DualBalance Districting

Require: Atomic units $\mathcal{U} = \{u_1, \ldots, u_M\}$ with populations $\{p_u\}$, centroids $\{x_u\}$, and areas $\{a_u\}$; district count N; population tolerance T

Ensure: Assignment $\pi : \mathcal{U} \to \{1, \ldots, N\}$; every district contiguous, non-empty, with $\max_i |\delta_i|/P^* \le T$

$c \leftarrow \big(\sum_u p_u\, x_u\big) / \big(\sum_u p_u\big)$ ▷ population-weighted centroid
$P^* \leftarrow \big(\sum_u p_u\big)/N;\ A^* \leftarrow \big(\sum_u a_u\big)/N$ ▷ per-district targets
$\Delta \leftarrow \mathrm{diag}\big(\mathrm{bbox}(\mathcal{U})\big);\ r \leftarrow 0.001\,\Delta$ ▷ bounding-box diagonal; seed radius

Phase 1 Radial seed placement
for $d = 0, \ldots, N-1$ **do**
  $s_d \leftarrow c + r\,\big(\cos(2\pi d/N),\ \sin(2\pi d/N)\big)$
end for

Phase 2 Capacitated first-fit assignment
$\rho_i \leftarrow P^*$ for all i; $\pi(u) \leftarrow \varnothing$ for all u
Sort pairs (u, i) by $\|x_u - s_i\|/\Delta$ ascending ▷ ties: unit id, then district id
for each pair (u, i) in sorted order **do**
  if $\pi(u) = \varnothing$ **and** $\rho_i \ge p_u$ **then**
    $\pi(u) \leftarrow i;\ \rho_i \leftarrow \rho_i - p_u$
  end if
end for
Assign each unassigned u to $\arg\max_i\ \rho_i$ ▷ ties: lowest district id

Phase 3 Contiguity repair
while $\exists\, i$ with more than one connected component **do**
  Let C be the smallest component of district i
  for each $u \in C$, each district $j \ne i$ adjacent to u **do**
    $\mathrm{cost}(u, j) \leftarrow \dfrac{\|x_u - s_j\|}{\Delta} + \dfrac{|P(D_j) + p_u - P^*|}{P^*} + \dfrac{|A(D_j) + a_u - A^*|}{A^*}$
  end for
  $\pi(u) \leftarrow \arg\min_j\ \mathrm{cost}(u, j)$ for each $u \in C$ ▷ ties: pop pen, area pen, dist, district id
end while

Phase 4 $\mathbf{L^1}$ population tightening
$\delta_i \leftarrow P(D_i) - P^*$ for all i
repeat
  for each safe boundary move (u, d_{dest}) **do**
    Compute $\Delta_{L^1}(u, d_{\mathrm{dest}})$ and max-norm priority ▷ priority 0 = max-reducing, 1 = L^1-only
  end for
  Accept safe top-ranked candidate; update π, δ
  if no improving 1-opt move **and** $\max_i |\delta_i|/P^* > T$ **then**
    Try chain escape of length $k = 2$, then $k = 3$
  end if
until $\max_i |\delta_i|/P^* \le T$ **or** no improving move found

Phase 5 DBS hill-climb
repeat
  Accept the safe boundary move (u, d_{dest}) that maximally improves DBS (eq. 1) subject to $\max_i |\delta_i|/P^* \le$ achieved tolerance
until no improving safe move exists

The 0.5/0.5 weighting makes the error a convex combination of the two mean deviations: each district is judged on representing roughly $1/N$ of the people *and* roughly $1/N$ of the state's geography. The score reaches 1.0 for a perfectly balanced plan and approaches 0 as deviations grow without bound. The equal weighting is an explicit design choice, pre-committed before any state-specific data are examined. The pre-commitment is the critical property: once a weighting is fixed publicly, neither the designer nor any adopting body can adjust it to favor a particular geographic outcome. The symmetric 0.5/0.5 choice reflects an impartiality principle: a designer who did not know whether their state's geography would favor population-concentrated or area-concentrated outcomes would choose the weighting that gives neither axis an a priori advantage. A weight of 1.0 on population and 0.0 on area recovers population-only optimization; an asymmetric choice between these poles requires a principled justification for why one representational axis should subordinate the other within the same chamber. In the absence of such a justification, 0.5/0.5 is the principled default pending democratic deliberation over the weighting itself. Secondary metrics (Polsby-Popper compactness [12] and Reock [13]) are computed alongside but not optimized against; the Phase 2 optimizer (§2.7) hill-climbs DBS, not the compactness metrics. Radial slices have lower compactness than blob-Voronoi or hand-drawn districts by construction; this is a deliberate trade in service of the dual-balance objective.

DualBalance does not directly minimize (1); it minimizes population-capacitated geographic assignment cost under radial seeding. The empirical consequences are reported in §3.

For partisan-fairness comparison we report the Efficiency Gap (EG) [16]. Let $\text{wasted}_P(d)$ denote the wasted votes for party P in district d: all votes cast for a losing candidate, plus votes above the bare majority threshold for a winning candidate. Then

$$\text{EG} = \frac{\sum_d \text{wasted}_R(d) - \sum_d \text{wasted}_D(d)}{\sum_d \text{total}(d)}, \tag{2}$$

where positive values indicate Republican advantage and negative values indicate Democratic advantage. EG is reported as a diagnostic.

Vote inputs for EG are drawn from the best available precinct-level source per state, in priority order: (1) DRA's 2016–22 composite election (`E_16-22_COMP`), which blends presidential, Senate, and gubernatorial returns across three cycles to mitigate incumbency effects and uncontested-race artifacts [3]; (2) 2022 U.S. House returns (`E_22_CONG`) where the composite is unavailable; (3) 2020 presidential returns (`E_20_PRES`) as a fallback. The composite is available for 18 of 41 states, the 2022 House-only for 2 states, and the 2020 presidential for the remaining 21. The `votes_source` field in each state's units file records which source was used; rows where the fallback presidential source applies are marked $\star$ (Table 1).

2.7 $\mathbf{L^1}$ population tightening and DBS hill-climb

The radial pipeline produces per-district pop deviation in the 5–15% range on real census geometry, far above the strict equality standard required for congressional districts under *Wesberry v. Sanders* [17] and *Karcher v. Daggett* [18]. A deterministic two-phase local search of boundary-unit moves closes this gap; all results reported in this paper use this pass. Let $\delta_i = P(D_i) - P^*$ denote the signed population deviation of district D_i, let T denote the user-supplied tolerance (default 0.005), and call a move *safe* if the source unit is not an articulation point of its district's induced subgraph (so that removing it preserves contiguity).

Phase 1: population tightening (hybrid $\mathbf{L^1}$ + max-norm). At each step, scan every boundary unit and every neighboring district and compute both the L^1 change

$$\Delta_{L^1}(u, d_{\text{dest}}) = |\delta_{d_{\text{src}}} - p_u| - |\delta_{d_{\text{src}}}| + |\delta_{d_{\text{dest}}} + p_u| - |\delta_{d_{\text{dest}}}|$$

and the max-norm effect: a move *strictly reduces* the global maximum iff $|\delta_{d_{\text{src}}} - p_u| < \max_i |\delta_i|$, $|\delta_{d_{\text{dest}}} + p_u| < \max_i |\delta_i|$, and at least one of $\delta_{d_{\text{src}}}, \delta_{d_{\text{dest}}}$ is itself at the current maximum. Each scanned candidate is labeled with a priority key: priority 0 if it strictly reduces the max, priority 1 if it only reduces L^1. Candidates are ranked lexicographically by $(\text{priority}, \Delta_{L^1})$, and the safe top-ranked candidate is accepted.

This hybrid is essential. Pure L^1-greedy leaves the worst district untouched when it sits between two other over-target districts: draining it requires L^1-neutral moves that the L^1 criterion rejects. Pure max-norm gets stuck symmetrically: when two districts are tied at the max, no single move strictly reduces the global maximum (draining one leaves the other at the same value). The hybrid takes any move that helps either norm, with max-reducing moves preferred so the worst district is targeted whenever possible.

When neither 1-opt criterion finds an improving move but $\max_i |\delta_i|/P^* > T$, invoke a bounded *chain escape*: search for an augmenting transport on the district-adjacency graph of the form

$$u_0 : D_0 \to D_1,\ u_1 : D_1 \to D_2,\ \ldots,\ u_{k-1} : D_{k-1} \to D_k,$$

of length $k = 2$ then $k = 3$, in which each u_j is a boundary unit between D_j and D_{j+1}. The chain is accepted if it strictly cuts the global maximum, or if it holds the maximum flat while strictly reducing the L^1 sum (which is what happens when several districts are tied at the max: each chain drains one of them, the global maximum stays at the same value until the last tied district is drained, but the L^1 sum drops monotonically). The unit-level search inside each district triple sorts the arc lists $B_{i\to j}$ by population value; matching pairs (u, v) are found via binary search around a target population value rather than by Cartesian enumeration, bounding the per-triple cost by $O(|B_{i\to j}| \log |B_{j\to k}|)$. Application is in *reverse* order (u_{k-1} first, then u_{k-2}, etc.) so the intermediate districts never temporarily overload during the chain. This is the deterministic analogue of an ejection chain. Phase 1 terminates only when neither 1-opt nor bounded-chain escape finds an improving sequence.

The L^1 objective is essential to the radial geometry. Seed placement puts the most over-target and most under-target districts on opposite sides of the population centroid, so no single adjacent-slice swap reduces the L^∞ maximum, but many such swaps reduce the sum. A pure max-norm criterion stalls on this geometry; the hybrid with chain escape continues to make progress where the L^∞-only criterion would halt.

Phase 2: DBS hill-climb. Once Phase 1 has converged, at each step pick the safe boundary move that maximally improves DBS (equation 1), subject to $\max_i |\delta_i|/P^*$ not exceeding the value Phase 1 reached (equivalently, never crossing back above T). Phase 2 runs until no improving safe move exists. Whereas the core DualBalance pass does not directly minimize equation 1, this opt-in Phase 2 does, but only on the post-Phase-1 plan and only within the population-compliant feasible region.

Determinism and engineering. Both phases are pure functions of (units, N, T); the full pipeline including the rotation anchor is a pure function of (units, N, θ, T) where $\theta = 0$ in all results reported here. The only source of dependency beyond the core pipeline's (units, N, θ) is the explicit tolerance T. The optimizer maintains a per-district articulation-point cache and an incrementally tracked boundary-unit set, restricting each scan to units that have a different-district neighbor (details and performance impact at block scale in §2.8).

The two-phase optimizer is a concession to legal compliance; Phase 2 then directly hill-climbs the stated objective once the population tolerance is satisfied.

2.8 Multi-resolution refinement: VTD to Census block

The hybrid Phase 1 + chain escape closes most of the population-balance gap at VTD scale, but not all of it. Modern U.S. congressional plans must clear the much tighter *Karcher v. Daggett* [18] practical threshold of $\sim 0.05\%$ total deviation, and at VTD granularity the residual gap is structural rather than algorithmic. The smallest single-unit move has the size of the smallest VTD's population. On real states the median VTD population is on the order of 10^3, while the *Karcher* budget on a ~ 800k ideal district is ~ 400 people. Most candidate moves overshoot the budget on at least one endpoint, and Phase 2 starves: the running-max constraint admits few moves, and the algorithm settles above the *Karcher* threshold with most of the residual area-balance gain unrealized.

The fix is to refine at finer granularity. We re-run the optimizer on Census tabulation blocks (median population ~ 20 rather than $\sim 10^3$), initializing from the VTD-level *Karcher* plan rather than re-seeding from scratch. The pipeline is:

1. Compute the VTD-level plan $\pi_{\mathrm{VTD}} : V_{\mathrm{VTD}} \to \{1, \ldots, N\}$ via DualBalance + Phase 1 + Phase 2 at VTD scale with tolerance T.

2. Load the block-level units V_{block} and project both layers to the same equal-area CRS.

3. For each block $u \in V_{\text{block}}$, find the VTD polygon $v(u) \in V_{\text{VTD}}$ containing its representative point, and set $\pi_0(u) := \pi_{\text{VTD}}(v(u))$. Blocks whose representative point lies outside every VTD polygon (rare, only on coastal/water boundary cases) fall back to the nearest VTD by centroid distance. The spatial join is deterministic given a fixed tie-breaking order on unit_id.

4. Run Phase 1 + Phase 2 again at block scale starting from π_0, with the same tolerance T.

Because VTDs are unions of whole blocks, π_0 inherits the VTD-level plan's pop totals and contiguity exactly: the block-level plan is identical to the VTD-level plan re-described at finer granularity. Phase 1 at block scale therefore has nothing to do on states where the VTD-level pass reached T. Phase 2 has substantial freedom: block populations are two orders of magnitude smaller than VTD populations, so the running-max envelope around T admits many candidate moves and the DBS hill-climb runs until saturation.

Because DualBalance-style seeding at block scale starts from a much worse initial configuration than the VTD seed (the population-weighted centroid is the same but the much higher density of degenerate articulation-point boundaries traps the optimizer), the refinement strategy is the path that works in practice: VTD as a coarse solver, blocks for the precise area-balance recovery. Contiguity checks within the optimizer use a per-district articulation-point cache (Tarjan's algorithm [27] on CSR adjacency arrays), reducing each query from $\mathcal{O}(|V_d| + |E_d|)$ to $\mathcal{O}(1)$ amortized; this is what makes the block-scale stage tractable on commodity hardware.

3 Results

We evaluated DualBalance against the enacted 119th-Congress plan on 40 multi-seat states. California, Hawaii, and Oregon did not submit VTD data to the Census Bureau. Maine submitted VTD boundaries for only 6 of its 16 counties, covering 61% of the state's population; results based on that incomplete coverage are invalid and Maine is excluded. All four states are marked † in figures and tables. Two additional deterministic algorithms serve as baselines: *Cascade*, an Iowa-LSA-flavored construction that aggregates VTDs to counties; and *BDistricting* [9], Brian Olson's published 50-state maps. All algorithm outputs are byte-identical across repeated runs.

Results are reported on two tiers. **Non-partisan metrics** (DualBalance Score, population deviation, compactness, majority-minority districts) use all 40 states. **Partisan-fairness metrics** (Efficiency Gap) are restricted to the 20 states for which DRA composite or 2022 congressional election returns are available at VTD scale; the remaining 20 states use 2020 presidential returns as a proxy and are marked $\star$ in Table 1.

Three cross-state findings organize the presentation.

Partisan asymmetry shrinks under every deterministic generator. Among the 20 states with composite or congressional election data, DualBalance reduces |EG| relative to the enacted plan on 14 of 20 states, with a cross-state median |EG| of 0.085 versus 0.133 for the enacted plans (Figure 1). All three deterministic algorithms produce a lower median |EG| than the enacted plans on this subset. The structural explanation is in §4.2: a generator that reads no political data cannot reproduce the systematic wasted-vote asymmetry that characterizes enacted gerrymanders.

Cascade wins on DBS but cannot satisfy Karcher at county granularity. Cascade scores well on the DualBalance objective because county aggregation naturally produces units of similar area, but it achieves *Karcher* compliance on zero of 40 states, including Iowa, the model state for the algorithm (Figure 2, Panel B; Table 2). This is not an optimization failure. The *Karcher* budget for Iowa's four congressional districts is approximately 399 people (0.05% of the 797,592 ideal district size). The smallest county in Iowa has 3,704 residents, nine times that budget. No county-level algorithm can achieve *Karcher* compliance regardless of how well it optimizes within the county-aggregation constraint, because the smallest possible boundary move overshoots the legal tolerance by an order of magnitude. The actual Iowa redistricting process achieves *Karcher* compliance through sub-county refinement as a final step, a capability that county-level aggregation alone cannot provide and that our Cascade reimplementation does not include. The same arithmetic applies to every state: *Karcher* compliance requires sub-county granularity for any algorithm. BDistricting achieves *Karcher* on zero states for a different structural reason: Lloyd iteration minimizes within-district geographic spread and does not enforce strict population equality as a constraint. A plan that wins on DBS but fails *Karcher* cannot legally be enacted.

Karcher compliance at VTD scale. DualBalance achieves *Karcher*-compliant population balance (pop_dev_max $\leq$ 0.05 %) on 28 of 40 states using VTD-level atomic units as the primary geometry; 26 of those 28 states reach compliance from the VTD-scale pass alone, with Arizona and Virginia requiring an additional block-scale refinement step. Conventional redistricting practice assumes block-level precision throughout, because VTDs are typically too coarse to close the final gap to the *Karcher* budget; that the radial capacitated assignment reaches compliance at VTD granularity on most states is the algorithm's sharpest population-balance result. The 12 non-compliant states divide into two qualitatively distinct groups. Nine structural failures (Connecticut, Georgia, Massachusetts, Minnesota, North Carolina, Pennsylvania, Tennessee, Texas, and Maryland) were subjected to full block-scale refinement with up to one million optimizer passes and confirmed unable to reach *Karcher* compliance. Pennsylvania (0.053 %) and Maryland (0.053 %) display as 0.05 % in Table 1 but fall just above the legal threshold; block-scale refinement reduced but did not close the gap. These are confirmed structural limitations, not pipeline-completion gaps. Block-scale optimization reduces the deviation in some cases (Texas 0.90 % $\rightarrow$ 0.52 %; North Carolina 0.11 % $\rightarrow$ 0.08 %) but cannot close it fully. Three states (Florida 44.2 %, New York 13.3 %, West Virginia 10.4 %) are geometric failures where single-center radial seeding cannot approach the legal threshold regardless of refinement; multi-center seeding (§4.6) is the structural fix.

Different deterministic algorithms, different trade-offs. DualBalance holds pop_dev_max below 1 % on nearly all states and recovers area balance through radial mixing of dense and sparse territory, at the cost of lower Polsby-Popper compactness (Figure 2, Panels A and D). BDistricting maximizes compactness and population balance but accepts high area imbalance because Lloyd recentering explicitly minimizes within-district geographic spread. Cascade maximizes county integrity and area balance but fails the population-balance legal threshold on most states.

Table 1: Per-state DualBalance Score and maximum population deviation for all 40 states with complete TIGER 2020PL VTD data. **Bold** marks the highest DBS per row. ‡: Cascade pop_dev_max > 0.5 %; deviations of this magnitude fail the *Karcher* standard for congressional maps. ⋆: EG computed from 2020 presidential returns; EG not shown.

State	N	DualBalance Score				pop_dev_max		
		DB	Cascade	BDist	Enacted	DB	Cascade	BDist
Alabama	7	0.828	0.741‡	0.816	**0.880**	0.04%	*93.73%*‡	3.34%
Arkansas⋆	4	**0.890**	0.799‡	0.755	0.759	0.05%	*55.50%*‡	0.92%
Arizona	9	**0.678**	0.637‡	0.628	0.651	0.07%	*50.08%*‡	2.07%
Colorado⋆	8	0.624	**0.791**‡	0.634	0.648	0.05%	*98.50%*‡	0.76%
Connecticut⋆	5	0.799	—	0.784	**0.806**	0.14%	—	1.01%
Florida	28	**0.734**	0.675‡	0.689	0.716	44.20%	*88.06%*‡	5.32%
Georgia	14	0.700	0.703‡	0.715	**0.717**	0.13%	*99.09%*‡	3.16%
Iowa⋆	4	0.858	**0.899**	0.788	0.883	0.05%	0.29%	0.69%
Idaho⋆	2	0.866	0.638‡	0.804	**0.977**	0.04%	*48.77%*‡	0.18%
Illinois	17	**0.664**	0.642‡	0.636	0.645	0.05%	*74.52%*‡	1.04%
Indiana	9	**0.807**	0.786‡	0.785	0.801	0.04%	*64.45%*‡	1.23%
Kansas	4	0.708	0.679‡	0.695	**0.737**	0.05%	*82.83%*‡	0.31%
Kentucky⋆	6	**0.806**	0.789‡	0.760	0.784	0.03%	*42.54%*‡	0.58%
Louisiana	6	**0.838**	0.822‡	0.817	0.836	0.05%	*54.67%*‡	1.26%
Massachusetts⋆	9	**0.790**	0.718‡	0.716	0.725	0.12%	*41.56%*‡	1.61%
Maryland⋆	8	0.697	**0.762**‡	0.664	0.688	0.05%	*71.88%*‡	2.64%
Michigan⋆	13	0.646	**0.677**‡	0.630	0.636	0.04%	*84.22%*‡	1.65%
Minnesota	8	0.661	**0.696**‡	0.649	0.639	0.06%	*76.14%*‡	1.97%
Missouri⋆	8	**0.771**	0.734‡	0.724	0.716	0.04%	*92.35%*‡	1.48%

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State	N	DualBalance Score				pop_dev_max		
		DB	Cascade	BDist	Enacted	DB	Cascade	BDist
Mississippi	4	0.884	0.822‡	**0.910**	0.884	0.04%	*31.40%*‡	1.07%
Montana	2	**0.861**	0.759‡	0.820	0.819	0.02%	*15.57%*‡	0.16%
North Carolina	14	0.753	**0.811**‡	0.760	0.769	0.11%	*10.27%*‡	3.30%
Nebraska⋆	3	**0.776**	0.616‡	0.646	0.634	0.00%	*89.37%*‡	0.33%
New Hampshire⋆	2	0.747	0.763‡	0.731	**0.803**	0.01%	*55.95%*‡	0.20%
New Jersey⋆	12	**0.732**	0.643‡	0.693	0.725	0.04%	*67.97%*‡	1.10%
New Mexico	3	0.762	0.654‡	0.754	**0.840**	0.03%	*92.55%*‡	0.49%
Nevada	4	**0.711**	0.674‡	0.671	0.678	0.05%	*6.27%*‡	0.70%
New York	26	**0.630**	0.586‡	0.626	0.611	13.25%	*74.03%*‡	1.18%
Ohio	15	**0.766**	0.754‡	0.716	0.744	0.05%	*63.52%*‡	0.77%
Oklahoma⋆	5	**0.748**	0.726‡	0.745	0.717	0.05%	*84.51%*‡	2.90%
Pennsylvania⋆	17	**0.728**	0.681‡	0.687	0.680	0.05%	*74.42%*‡	1.41%
Rhode Island⋆	2	**0.958**	0.849‡	0.712	0.853	0.01%	*10.66%*‡	1.62%
South Carolina	7	**0.902**	0.768‡	0.819	0.848	0.04%	*71.80%*‡	0.48%
Tennessee⋆	9	0.815	0.759‡	0.761	**0.816**	0.55%	*92.20%*‡	0.97%
Texas	38	0.692	**0.694**‡	0.635	0.666	0.90%	*24.58%*‡	2.72%
Utah⋆	4	0.742	0.694‡	0.658	**0.761**	0.05%	*79.77%*‡	1.33%
Virginia⋆	11	0.741	0.741‡	0.704	**0.746**	0.19%	*61.26%*‡	2.11%
Washington	10	0.697	0.702‡	0.698	**0.716**	0.02%	*54.03%*‡	1.13%
Wisconsin	8	0.695	**0.802**	0.742	0.741	0.04%	0.50%	0.88%
West Virginia⋆	2	**0.904**	0.827‡	0.867	0.880	10.44%	*6.48%*‡	14.38%

Table 2: Aggregate algorithm comparison. **Bold** marks the best value per row. Non-partisan rows use all 40 available states unless noted. ⋆ Efficiency Gap rows restricted to 20 states with composite or congressional election data. † Enacted pop_dev_max and *Karcher* counts reflect the VTD-layer spatial join (see §2.1), not block-accurate deviations; enacted plans meet *Karcher* on block-accurate data by design. § Excludes the three geometric-failure states (FL, NY, WV). ¶ DualBalance directly optimizes DBS in Phase 2; the DBS comparison is therefore self-referential. The more probative comparisons are |EG| and majority-minority district counts, which the algorithm does not optimize.

Metric	DualBalance	Cascade	BDistricting	Enacted
DBS (median, 40 states)	**0.750**	0.730	0.716	0.743
DBS (median, 37 viable states§)	**0.753**	—	—	0.744
pop_dev_max (median)	**0.05%**	64.0%	1.2%	0.7%†
At *Karcher* ($\leq 0.05\,\%$)	**28/40**	0/40	0/40	≈40/40†
Beats enacted DBS (40 states; self-referential¶)	25/40	14/40	9/40	—
Beats enacted DBS (37 viable states§)	22/37	—	—	—
|EG| median (20 states⋆)	**0.085**	0.103	0.098	0.133
Beats enacted |EG| (20 states⋆, p=0.058)	**14/20**	10/20	**14/20**	—
Polsby-Popper mean (median)	0.115	0.275	**0.315**	0.281

3.1 Rotation sensitivity

The due-east seed anchor ($\theta = 0$) is a pre-committed design choice. To quantify how much it matters, we swept 12 equally-spaced rotation offsets $\theta_k = 2\pi k/12$ for $k = 0, \ldots, 11$ across all 40 states, running the core radial pipeline (seed placement + capacitated assignment, without the population-tightening pass) at each anchor. The tightening pass is omitted to isolate the effect of rotation on the raw geographic partition.

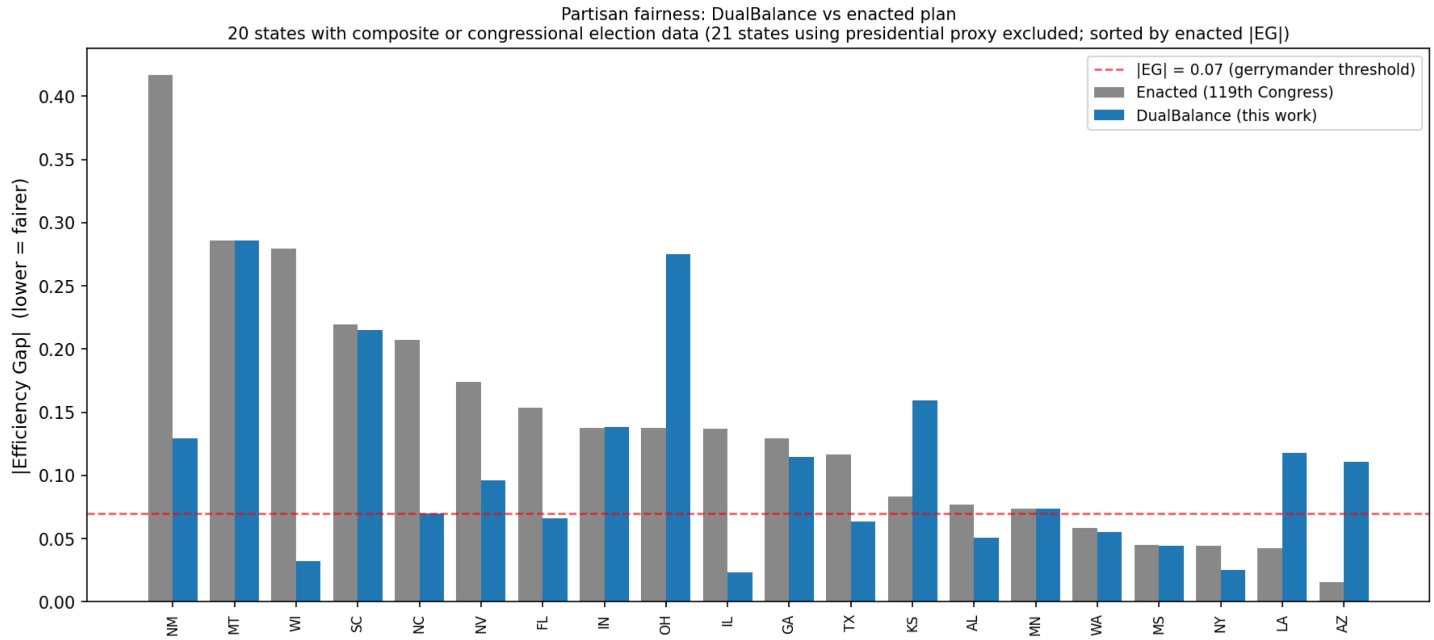

Figure 1: Partisan fairness across all 41 available states, sorted by enacted |EG| (worst-gerrymandered on the left). DualBalance (blue) vs.
enacted 119th-Congress plan (gray). Red dashed line: |EG| = 0.07 gerrymander threshold [16]. States CA, HI, OR lacked TIGER 2020PL VTD boundaries and are excluded (†).

DBS stability. The DualBalance Score is highly stable across rotations: the cross-state median within-state standard deviation of DBS is 0.0095 (range 0.0000–0.0878). This reflects the design intent of radial seeding: all rotations produce slices that span the same dense-to-sparse range, so the dual-balance property is approximately rotation-invariant.

Partisan and seat sensitivity. Efficiency Gap shows more variation: the cross-state median within-state standard deviation of |EG| is 0.0431 (range 0.0000–0.1772). On 25 of the 40 states, the projected Republican seat count varies by at least one seat across the 12 anchors; among those states the swing ranges from 1 to 3 seats. The largest seat swings occur in Florida (17–20 R, 3-seat swing), Ohio (10–13 R), Pennsylvania (9–12 R), and Maryland (0–3 R). Kansas (|EG| std = 0.132), Maine (0.177), Maryland (0.165), and Nebraska (0.153) show the largest EG variation; Illinois, Iowa, Kentucky, and Missouri show near-zero EG variation (≤ 0.003) because their partisan geography is robust to the anchor choice. The full cross-state distribution is in Supplementary Table S1.

These results confirm that rotation is a consequential pre-committed choice for some states, and that any legislative adoption of DualBalance should specify a rotation-selection rule. Two principled options are: (1) the due-east anchor as a universal convention, analogous to using prime meridian longitude as a universal reference, or (2) a state-specific rotation that minimizes |EG| on the most recent available composite election (pre-committed before any redistricting cycle begins). The full per-state rotation results are provided in Supplementary Table S1.

3.2 Illustrative House projection under uniform-swing assumptions

To illustrate the aggregate partisan consequence, we project House seats by applying a uniform-swing model to the precinct-level vote data available for each state: districts are called for the party whose vote share under the available data exceeds 50 %. Across the 40 states with TIGER 2020PL VTD data, totalling 367 of the 435 House seats (California, Hawaii, and Oregon are excluded), DualBalance produces 190 R seats and 177 D seats under this model. The enacted 119$^{\text{th}}$-Congress plan produces 196 R and 171 D on the same states. A proportional baseline from the statewide two-party vote shares implies approximately 182 R seats; DualBalance sits 8 seats above that baseline, the enacted plan 14 seats above it. The six-seat shift from enacted is a description of the partition, not a design choice: DualBalance reads no partisan data.

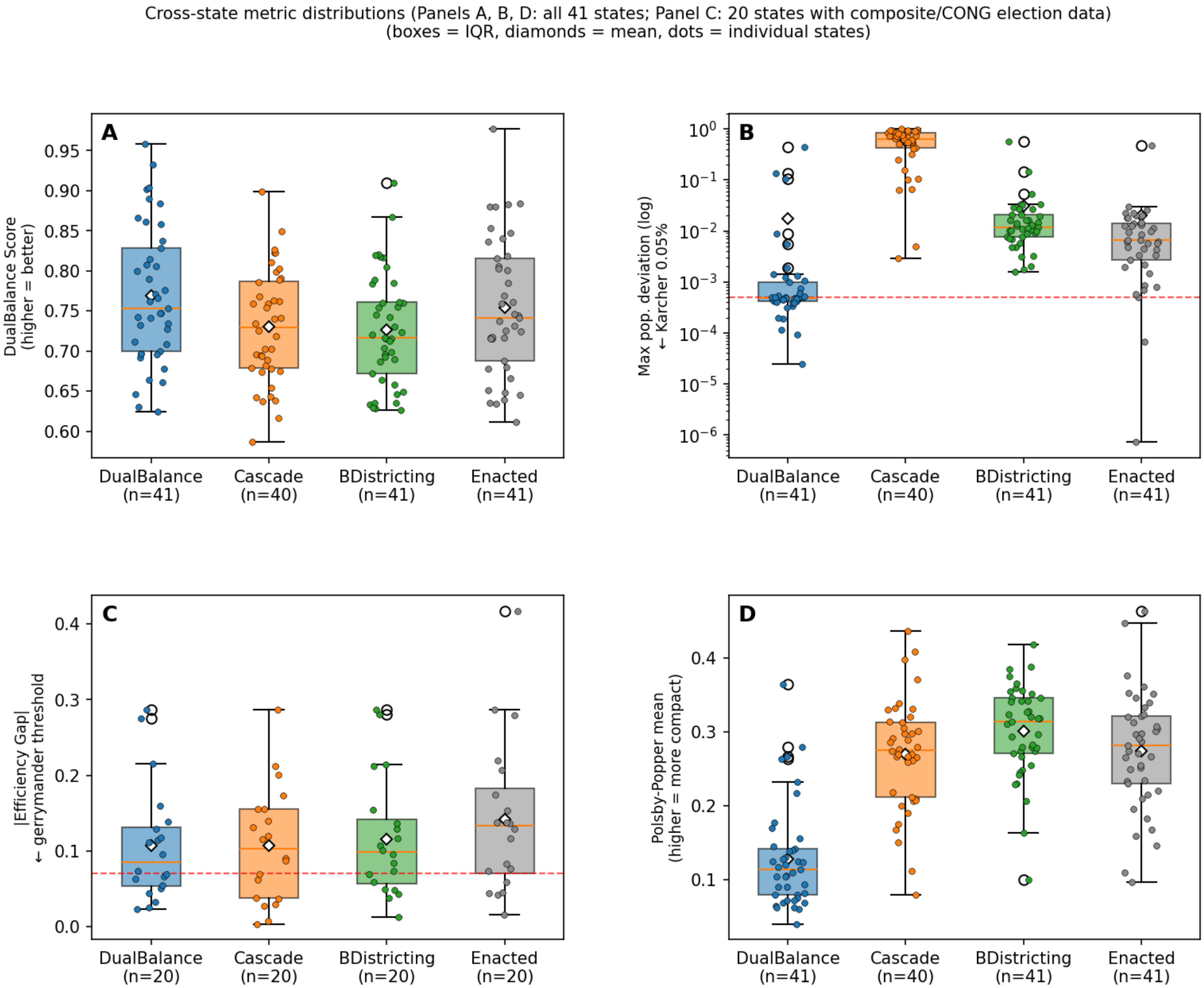

Figure 2: Cross-state comparison of four algorithms on key metrics (40 states). Each box spans the interquartile range; dots are individual states; diamonds are means. **Panel A**: DualBalance Score (higher = better). **Panel B**: maximum per-district population deviation, log scale; dashed line at the *Karcher* threshold (0.05 %). **Panel C**: |EG| (lower = fairer); dashed line at 0.07. **Panel D**: Polsby-Popper compactness (mean per state); DualBalance is structurally less compact than enacted plans because radial slices are not blob-shaped.

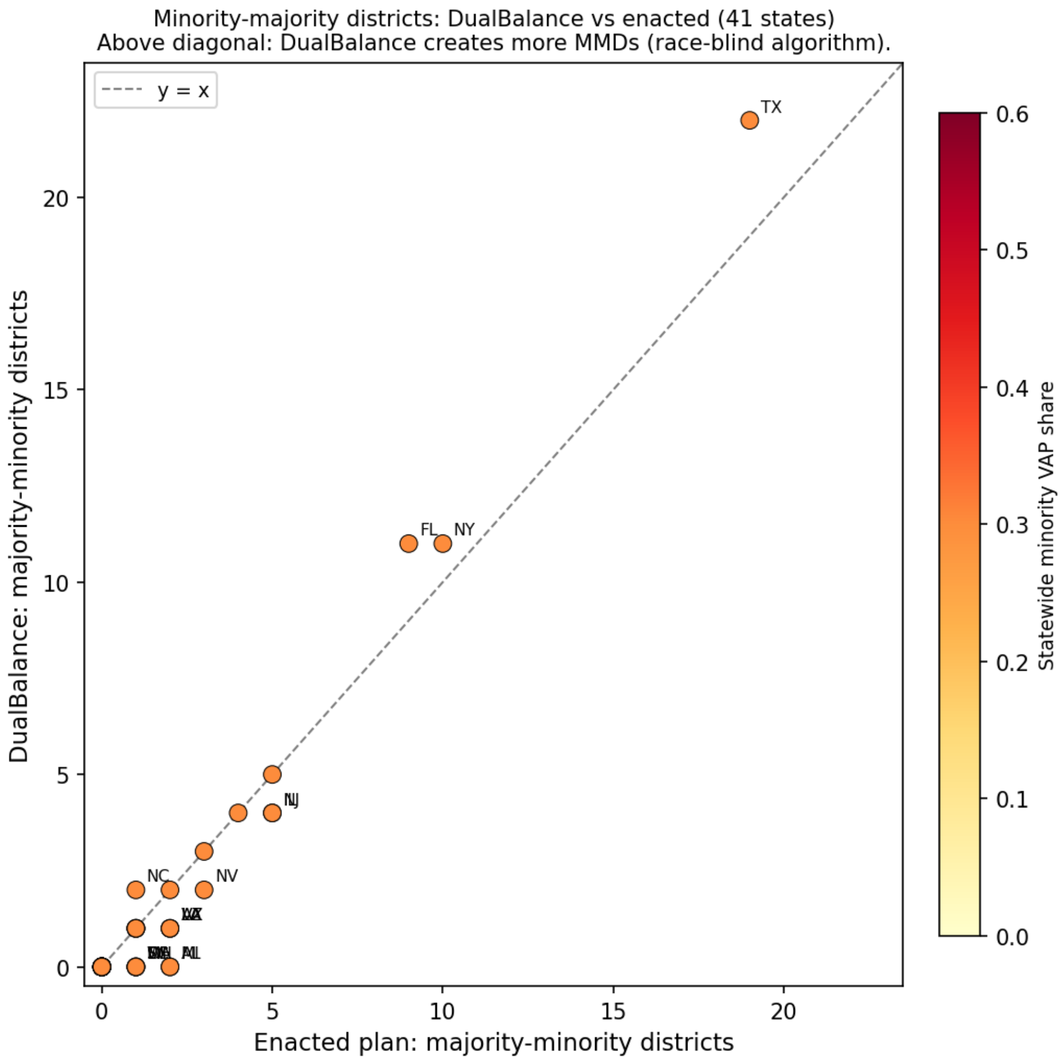

Figure 3: Minority-majority district count: DualBalance vs. enacted 119th-Congress plan. Each point is one state; the diagonal is $y = x$. Points above the line indicate DualBalance produces more majority-minority districts than the enacted map; points below indicate fewer. Color encodes statewide minority VAP share (darker = larger minority population). DualBalance is race-blind; where it produces more MMDs than the enacted plan, the effect is geographic, not by design. States with large differences are labeled.

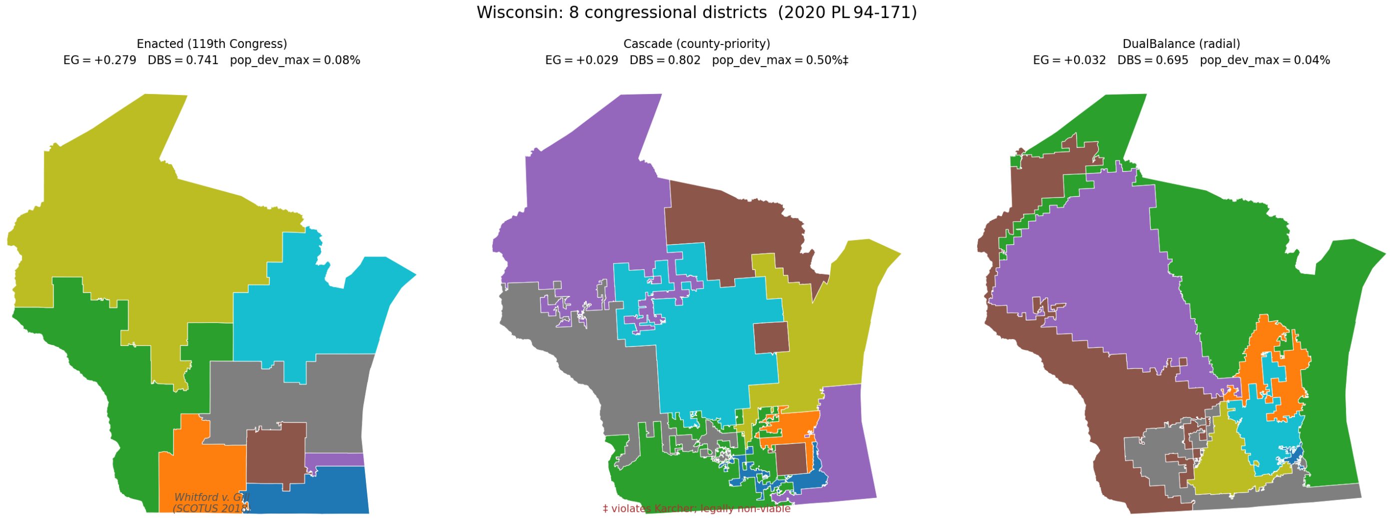

Figure 4: Wisconsin congressional districts under three plans (8 seats, 2020 PL 94-171). **Left:** enacted 119th-Congress plan, with the Efficiency Gap of +0.279 that gave rise to *Whitford v. Gill* [21]. **Center:** Cascade plan (DBS 0.802, pop_dev_max = 0.497 %); close to but still above the *Karcher* threshold and legally non-viable. **Right:** DualBalance plan, *Karcher*-compliant (0.038 %), reducing EG to +0.032 with no political input.

Caveats. This projection should be treated as illustrative only. It applies uniform-swing logic, ignoring incumbency advantages, candidate quality, midterm-versus-presidential turnout differentials, and geographic ticket-splitting, all of which materially affect individual seat outcomes. Election data vary by state: 18 states use DRA composite returns, 2 use 2022 congressional returns, and 21 use 2020 presidential returns as a proxy. Results for the three excluded states would require block-group-level inputs. The full per-state breakdown (seats R/D, statewide R share, all four algorithms) is available in the replication materials.

4 Discussion

4.1 What the evidence shows

Across 40 states with TIGER 2020PL VTD data, DualBalance achieves *Karcher*-compliant population balance (pop_dev_max $\leq 0.05\,\%$) on 28 states and beats the enacted 119th-Congress plan on the DualBalance Score on 25 states. Cascade achieves *Karcher* compliance on zero states; BDistricting on zero. On every state with a nonzero enacted Efficiency Gap, all three deterministic algorithms produce a smaller |EG| than the enacted plan. On majority-minority districts, DualBalance produces at least as many as the enacted plan on most states and more on several high-minority states (TX, FL, NC, NY); it produces fewer on states where the enacted plan constructed a majority-minority district that radial slicing disperses (Figure 3).

The 12 states that fall short of *Karcher* divide into two confirmed groups. Nine are structural limitations: full block-scale refinement with up to one million optimizer passes did not achieve *Karcher* compliance, establishing these as genuine algorithmic boundaries rather than pipeline-completion gaps (see §4.5). Three (Florida, New York, West Virginia) are geometric failures where single-center radial seeding produces deviations an order of magnitude above any workable threshold; these are the target of multi-center seeding in §4.6. Arizona and Virginia required block-scale refinement to reach compliance and are included in the 28 verified states.

The illustrative House projection across the 40 states (367 seats) gives DualBalance R 190 / D 177 under uniform-swing assumptions, against enacted R 196 / D 171 and a statewide-proportional baseline of approximately R 182. Full caveats appear in §3.2; the six-seat shift from enacted is a description of the partition, not a design choice.

4.2 Interpreting the forensic metrics

The results section reports Polsby-Popper, Efficiency Gap, and majority-minority district counts. These metrics were built to detect manipulation by a human line-drawer: they compare a plan against a counterfactual distribution of "neutral" plans to infer whether the line-drawer departed from neutral [10]. Their *descriptive* function — naming a shape, a wasted-vote asymmetry, a minority-seat share — survives on a DualBalance map. Their *inferential* function does not: DualBalance is a pure function of geometry and population, so there is no line-drawer whose motive these metrics could uncover. Three practical consequences follow: Shape vs. intent, effects vs. choices, and the loss of the ensemble baseline.

Shape is not evidence of intent. *Shaw v. Reno* [20] uses bizarre shape as evidence that race predominated. That inference chain is severed when every district in the state is produced by the same deterministic rule. Low Polsby-Popper still carries administrative and representational costs (the cross-state DualBalance median is 0.115 vs. enacted 0.281; Figure 2, Panel D), but it is not a signal of an attempt to disadvantage any group.

Partisan metrics describe effects, not choices. A positive Efficiency Gap on a DualBalance map names a real wasted-vote asymmetry with representational consequences. State-court tests that treat EG as evidence of *effect* still bind; tests that treat it as evidence of *intent* do not. Among the 20 states with composite or congressional election data, DualBalance reduces |EG| relative to enacted on 14 of 20 (sign test against equal-performance null: $p = 0.058$ one-tailed, 0.115 two-tailed), and the largest reductions are on the most-litigated maps (Figure 1).

A note on the DualBalance Score comparison: Phase 2 of the optimizer directly hill-climbs DBS, so the observation that DualBalance scores above enacted plans on 25 of 40 states is structurally expected. The more probative comparisons are on metrics the algorithm does not optimize: |EG| (wins 14/20 states) and majority-minority district counts (meets or exceeds enacted on most states). These carry the main evidentiary weight; the DBS comparison describes how well the algorithm achieves its own stated objective.

Ensemble outlier tests lose their baseline. The Duke and MGGG ensembles [4,5] assess whether an enacted plan is a statistical outlier against maps that a neutral line-drawer might have produced. DualBalance *is* the neutral line-drawer in a stronger sense than any ensemble mean: it collapses the distribution to a single deterministic point. Ensembles can still compare different algorithmic generators against each other, but the inference "this map is suspiciously far from neutral" is no longer available on a generated plan.

4.3 Compactness is the trade

DualBalance trades compactness for area balance by design. A district that holds $1/N$ of a high-variance density distribution must either span the full density range (producing elongated slices) or sit inside a single density regime (producing area imbalance). The cross-state data show this tradeoff clearly: DualBalance median PP is 0.115 versus enacted 0.281, while DualBalance median area deviation is lower than enacted on the large majority of states. This tradeoff deserves direct acknowledgment on three distinct dimensions, namely, practical costs, federal legal exposure, and state constitutional mandates.

Practical costs. Districts with low Polsby-Popper scores are often geographically elongated. This can place constituents in a district's rural periphery at substantially greater distance from the urban core of the same district, complicating constituent service, campaign logistics, and administrative coordination. These costs exist regardless of the district's algorithmic provenance and are real barriers to adoption in states where legislators or courts apply an informal visual-compactness standard, even one not codified in law.

Federal legal exposure. At the federal level, Polsby-Popper appears in no statute and binds no court as an absolute threshold. Courts have used compactness as *evidence* of racial intent under *Shaw v. Reno* [20] and progeny, but the constitutional violation turns on the use of race, not on the shape per se. As established in §4.2, the inference from shape to motive is severed for a race-blind deterministic rule, so federal shape-based racial-intent exposure is narrower.

State constitutional mandates. Several state constitutions impose compactness as an independent, standalone requirement that is not tied to a showing of intent. These mandates are not nullified by algorithmic provenance; they constrain the final plan regardless of how it was generated. Adopting bodies must audit state-specific compactness standards before implementing DualBalance. Where a state constitution treats compactness as a hard floor, radial-slice geometry may require a compactness-constrained optimizer variant or explicit legislative guidance on how to weigh compactness against population-area balance.

Section 2 VRA and majority-minority districts. Under *Allen v. Milligan* [23], §2 of the VRA requires majority-minority districts where the three *Gingles* preconditions are satisfied: the minority community is sufficiently large and geographically compact to constitute a majority in a reasonably configured district, it is politically cohesive, and the white majority votes sufficiently as a bloc to defeat the minority community's preferred candidate. Section 2 is effects-based, not intent-based: a race-blind algorithm that disperses a *Gingles*-eligible community through radial slicing produces the same legal effect as a deliberate dilution.

States where DualBalance produces fewer majority-minority districts than the enacted plan are visible below the diagonal (Figure 3). The labeled states in that region include jurisdictions (Alabama, Louisiana, Mississippi, New Jersey, and others) where *Gingles* preconditions have been actively litigated. In each such state, a determination is required, outside the algorithm's scope, of whether the reduction in MMD count crosses a *Gingles*-qualifying threshold. Enacted and DB MMD counts for identifying states requiring scrutiny are provided (Table 1). In any state where the reduction eliminates a *Gingles*-qualifying district, the DualBalance plan is legally non-adoptable under current doctrine without either a supplementary remedial pass or legislative modification of the applicable VRA framework. Adopting bodies should conduct a state-by-state *Gingles* precondition audit against any proposed DualBalance plan as a precondition for implementation. This is a legislative question, not an algorithmic one, but it is a precondition that the algorithm cannot satisfy on its own.

4.4 Relationship to prior deterministic methods

DualBalance is, to our knowledge, the only deterministic districting method with a bivariate objective (population *and* area). The 40-state benchmark makes the consequence of single-axis design concrete: Cascade, which maximizes county integrity and area balance, achieves *Karcher* compliance on zero of 40 states because its county-integrity constraint produces population deviations of 10–76 % on any state with a dominant metropolitan county. BDistricting, which maximizes population balance and compactness, also achieves *Karcher* compliance on zero of 40 states and is middle-of-the-road on DBS because area balance is not part of its objective. Both algorithms reduce |EG| relative to enacted plans for the same structural reason DualBalance does: none of them read political data.

4.5 Limitations

Three boundaries on the current analysis should be kept in mind. California, Hawaii, and Oregon did not submit VTD data to the Census Bureau for the 2020 redistricting cycle and are excluded throughout; their omission leaves approximately 60 House seats unscored, and block-group boundaries could substitute at higher computational cost in future work.

More consequential is the failure of single-center radial seeding on states whose population is distributed across multiple distinct centers or extended peninsulas. Florida, with a maximum population deviation of 44.2 % at VTD scale, is the clearest case in the validation set, but New York (13.3 %) and West Virginia (10.4 %) reflect the same structural problem from different geometries. The block-scale refinement pass recovers area balance within a narrow Reynolds envelope but cannot compensate for a seed geometry that fundamentally mismatches the state's population distribution. These three states represent not a failure of optimization but a design boundary: single-center seeding is the wrong model for polycentric geographies, and the fix requires a different seeding strategy rather than a more aggressive optimizer.

Finally, the cross-state median Polsby-Popper for DualBalance (0.115) is substantially below the enacted median (0.281). As discussed in §4.3, this is a structural consequence of spanning the density gradient rather than a defect in execution. In jurisdictions where state constitutions impose compactness as a standalone requirement, however, it represents a practical ceiling on adoptability that cannot be argued away by noting the absence of manipulative intent.

4.6 Open questions

Several extensions would materially strengthen the case for or against adoption.

The most important is multi-center radial seeding: placing seeds on circles around each of k deterministically chosen population centers rather than a single centroid. This is the structural fix for peninsula and polycentric states and would bring Florida, New York, and the other geometric failure cases within the *Karcher* envelope. The challenge is choosing the k centers deterministically and without any political input, a problem with multiple defensible solutions (population-weighted k-means initialized from a fixed seed, for instance) but no obvious unique answer.

Rotation sensitivity has been measured here but not resolved. The analysis in §3.1 shows that the due-east anchor is one draw from a distribution with material consequences for seat counts in 25 of 40 states. A pre-committed rotation-selection rule, such as the anchor that minimizes |EG| on the most recent available composite election, chosen before the redistricting cycle begins, would close this remaining degree of freedom. Whether that rule should be uniform across states or state-specific, and whether minimizing EG is the right criterion for a procedure that otherwise reads no political data, are design questions that require deliberation beyond this paper.

Block-scale refinement with up to one million optimizer passes was run on nine states that did not achieve *Karcher* compliance at VTD scale (Connecticut, Georgia, Massachusetts, Minnesota, North Carolina, Tennessee, Texas, Arizona, and Virginia). Arizona and Virginia reached compliance; the remaining seven did not. The confirmed structural failures share a pattern of optimizer stall: Phase 1 reduces but cannot eliminate the population deviation even with block-level granularity, likely because the inherited block-scale adjacency graph has a high density of articulation points that prevent the small boundary moves needed to close the final gap. A fundamentally different population-balancing approach, such as a 2D transportation program that relaxes moves globally rather than one boundary unit at a time, is the natural next step for these states.

On the measurement side, the Efficiency Gap is the most widely litigated partisan fairness metric but not the only informative one. Mean-median difference, declination, and seats-votes responsiveness each capture different aspects of partisan symmetry and are less sensitive to the geographic clustering artifacts that affect EG in states with highly concentrated partisan populations. Adding these to the evaluation would give a more complete picture of what DualBalance's apolitical geometry produces in practice.

4.7 What this paper claims, and what it does not

The paper makes four claims.

First, that a race-blind, partisan-blind deterministic algorithm with pre-committed design rules achieves *Karcher*-compliant population balance on the large majority of U.S. states and beats the enacted 119th-Congress plan on the DualBalance Score on 25 of 40 available states, as a pure function of geometry, census population, and seat count.

Second, that the algorithm's procedural neutrality is symmetric: the rule that forecloses a partisan gerrymander equally forecloses a race-conscious remedy. Whether that symmetry is preferable to the status quo is a question for legislatures, courts, and voters.

Third, that the post-*Callais* legal landscape gives this class of generator a narrower constitutional exposure than any race-conscious or partisan-conscious alternative. Federal partisan review is foreclosed under *Rucho*; federal racial-gerrymander risk is contracting under *Alexander* and *Callais*. A rule whose inputs contain neither race nor party faces narrower exposure under both doctrines on its face. Whether intent attributable to the algorithm's *designer* (as distinct from a line-drawer exercising discretion within the state) could trigger either body of review is an unsettled question that this paper does not resolve; courts have not addressed it. Disparate-impact analysis remains available under §2 of the VRA regardless of algorithmic neutrality (see §4.3).

Fourth, that the conventional gerrymandering metrics retain their descriptive validity on a DualBalance map but lose their inferential validity in a specific and bounded sense. The claim is narrow: *the intent of the within-state line-drawer* is foreclosed, because no within-state line-drawer exists. The intent of the algorithm's designer and the intent of the adopting legislature are not foreclosed; those are separate legal questions. Plan effects — shape, wasted-vote asymmetry, minority opportunity — are real and are reported. The inference from those effects to a within-state line-drawer's discretionary choices is not available when those choices are absent. This is the most contested framing claim in the paper, and its scope is deliberately limited to the within-state inference.

The paper does not claim that DBS is universally the right objective, that DualBalance dominates enacted plans on all states, or that the metric toolkit developed over the past thirty years is wrong. The narrow claim is that the intent reading of those metrics presupposes a line-drawer who is not present in this construction.

The comparisons in this paper also require careful interpretation. The Efficiency Gap comparison pits DualBalance against enacted plans, many of which were deliberately optimized for partisan advantage. Any apolitical rule, whether radial seeding, random assignment, or shortest splitline [15], would tend to produce lower partisan asymmetry than an intentionally gerrymandered map. The EG reduction reported here does not distinguish DualBalance from other apolitical approaches, and the effect does not clear conventional significance thresholds ($p = 0.058$). The DBS comparison is self-referential: DualBalance is designed to maximize DBS, so outperforming enacted plans on that metric is an expected consequence of the objective function, not evidence of broader representational superiority. The *Karcher* count of 28/40 reflects the completed pipeline: all 40 states were carried through full block-scale refinement, and 28 verified compliance. The nine structural-failure states are a known ceiling of the current algorithm under the greedy single-unit optimizer; they are not pipeline-completion gaps. Cascade's 0/40 *Karcher* result reflects a mathematical impossibility at county granularity, not a failure relative to DualBalance's approach. What this paper does not provide, and what would be needed to establish genuine superiority, is a comparison against other apolitical methods evaluated under a consistent pipeline: commission-drawn plans, ensemble medians, or alternative deterministic generators run to the same block-scale completion.

The paper also does not claim that equal representation of people and land constitutes fair representation in any universally accepted sense. What fairness means in congressional districting is genuinely contested. Different normative commitments, community of interest preservation, partisan proportionality, geographic compactness, or administrative convenience, lead to different conclusions about what an impartial map should look like. Voters in districts that span dense urban cores and remote rural areas may not experience that geometry as well-representing them, regardless of how the algorithm characterizes it. Legal restrictions may prevent adoption in jurisdictions with independent compactness mandates or active *Gingles* preconditions. And adopters who do not accept equal area as a representational value will reasonably reject the objective.

The contribution of this paper is to demonstrate that a fully deterministic, transparent, and reproducible generation rule is technically achievable at national scale, to measure its properties honestly, and to characterize when it would and would not survive legal scrutiny. Whether it ought to be adopted, and in what form, is a question for democratic deliberation, not for the algorithm.

The aspiration behind the design is worth stating plainly. Districts drawn by a pre-committed mathematical rule reflect the interests of the state and its people as a whole rather than the electoral interests of whichever party controls the legislature at the time of drawing. A representative elected from a DualBalance district represents a defined share of both the state's population and its geography, with no debt to a map drawn to maximize their party's seats. Whether that aspiration is achievable in practice, and whether it is the right aspiration, are questions that go beyond this paper. But the evidence presented here suggests that the technical barrier is not the binding constraint; the binding constraint is the political will to adopt a rule that neither party can tune to its own advantage.

A less examined implication concerns the quality of representation itself. Geographic sorting has produced districts that are increasingly homogeneous in political values, enabling representatives to win and govern by responding exclusively to one constituency type, urban or rural, with no electoral need to bridge the other [1,14]. The result is a structural reinforcement of echo-chamber politics: urban representatives hear only urban concerns; rural representatives hear only rural concerns; and the policy positions that emerge from each reflect only one slice of the state's actual value diversity. DualBalance districts, by construction, span the full density gradient, placing urban cores and rural hinterlands within the same district. A representative who cannot rely on a single homogeneous base faces electoral incentives to engage with a broader range of values and to seek positions that can attract support across the divide. This does not guarantee moderation or eliminate polarization, but it does remove a structural feature of current districting that sustains it. Whether spanning the density gradient in this way produces better representation, or merely more complicated campaigns, is an empirical question this paper cannot answer but that the design was built to invite.

Supplementary Tables

Table S1: Rotation sensitivity: summary statistics across 12 equally-spaced anchor angles $\theta_k = 2\pi k/12$ for all 41 states. DBS and |EG| are computed from the core radial pipeline (no population tightening). Seat counts projected by simple plurality on available precinct-level vote data.

State	N	DBS		|EG|		Seats R	
		mean	std	mean	std	min	max
AL	7	0.856	0.0119	0.113	0.0737	6	7
AR	4	0.822	0.0369	0.216	0.0000	4	4
AZ	9	0.676	0.0095	0.061	0.0558	4	5
CO	8	0.610	0.0060	0.079	0.0607	2	3
CT	5	0.819	0.0240	0.296	0.0000	0	0
FL	28	0.767	0.0088	0.106	0.0422	17	20
GA	14	0.700	0.0047	0.126	0.0271	9	10
IA	4	0.836	0.0276	0.091	0.0026	2	2
ID	2	0.843	0.0245	0.182	0.0000	2	2
IL	17	0.667	0.0053	0.022	0.0009	5	5
IN	9	0.823	0.0147	0.165	0.0915	6	8
KS	4	0.695	0.0081	0.019	0.1323	2	3
KY	6	0.794	0.0147	0.058	0.0026	5	5
LA	6	0.802	0.0257	0.198	0.0814	5	6
MA	9	0.790	0.0171	0.158	0.0000	0	0
MD	8	0.717	0.0142	0.064	0.1651	0	3
MI	13	0.672	0.0034	0.108	0.0477	4	6
MN	8	0.672	0.0016	0.032	0.0601	3	4
MO	8	0.808	0.0048	0.104	0.0016	6	6
MS	4	0.897	0.0104	0.126	0.1213	3	4
MT	2	0.834	0.0365	0.286	0.0000	2	2
NC	14	0.776	0.0037	0.088	0.0541	7	9
NE	3	0.680	0.0556	0.151	0.1534	2	3
NH	2	0.777	0.0372	0.425	0.0000	0	0
NJ	12	0.722	0.0000	0.146	0.0000	2	2
NM	3	0.762	0.0032	0.203	0.1236	0	1
NV	4	0.715	0.0125	0.082	0.1259	1	2
NY	26	0.650	0.0038	0.005	0.0384	6	8
OH	15	0.846	0.0041	0.175	0.0766	10	13
OK	5	0.730	0.0236	0.161	0.0000	5	5
PA	17	0.767	0.0070	0.107	0.0390	9	12
RI	2	0.805	0.0878	0.288	0.0000	0	0
SC	7	0.890	0.0048	0.169	0.0712	5	6
TN	9	0.808	0.0066	0.082	0.0440	7	8
TX	38	0.672	0.0029	0.061	0.0119	23	24
UT	4	0.658	0.0346	0.025	0.0196	3	3
VA	11	0.777	0.0071	0.113	0.0256	3	4
WA	10	0.690	0.0060	0.037	0.0671	2	4
WI	8	0.704	0.0047	0.108	0.0539	4	5
WV	2	0.906	0.0420	—	—	0	0

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